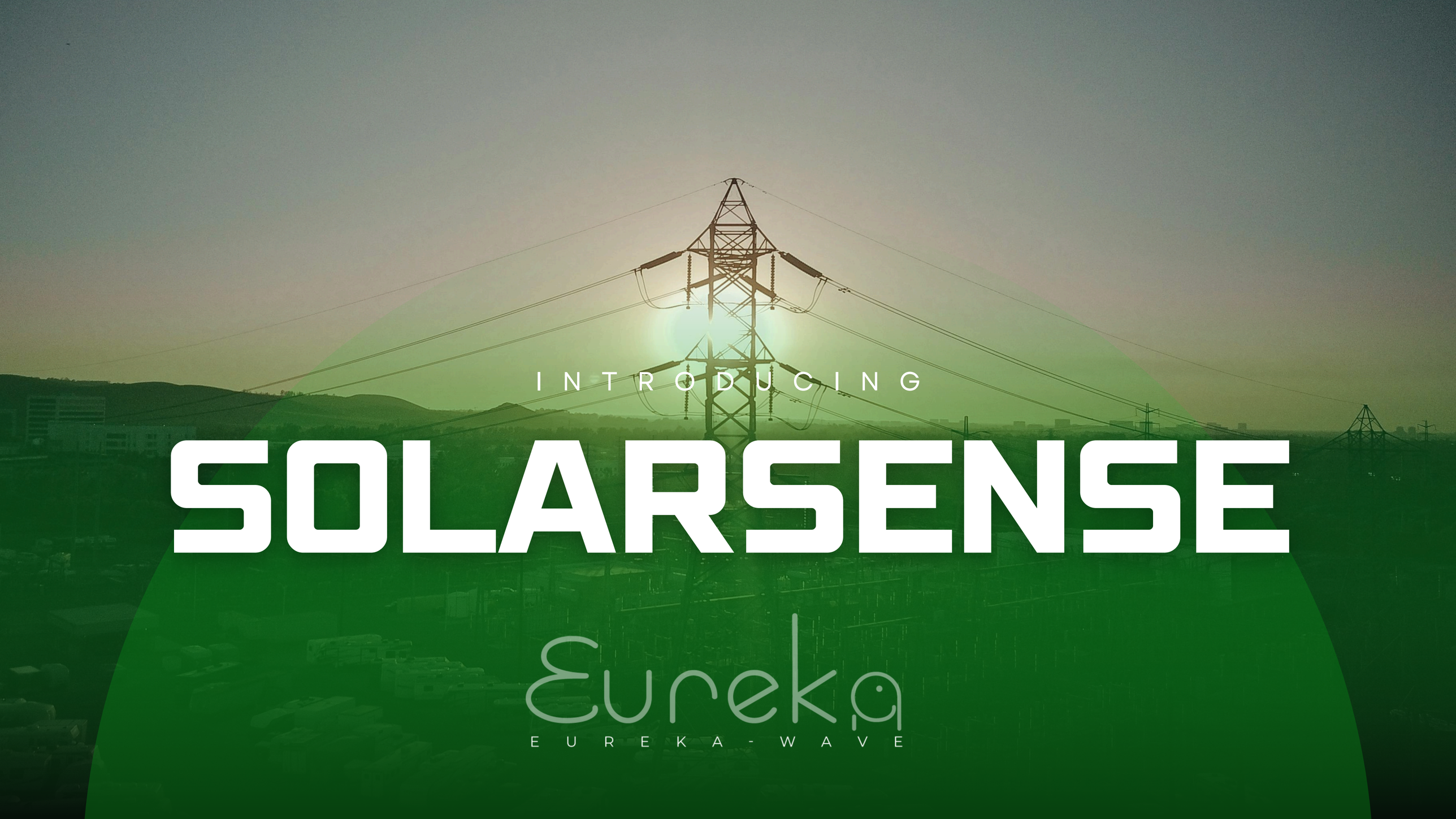

INTRODUCING
SOLARSENSE
Eureka
EUREKA-WAVE

EUREKA
WAVE



15%

Efficiency loss due to improper placement

30%

Increased cost from inaccurate sizing

WHY SOLARSENSE?

- Solar investments often rely on guesswork
- Current tools:

Satellite data only
Software simulations only
Expensive hardware

- Micro-climate variations- make predictions unreliable
- Uncertainty about investing in solar panels.
- No accurate measurements
- More EVs
- Good data - smarter, cheaper energy decisions.
- Lack of accessible, local, on-site irradiance data for Sri Lanka
- Decision-makers need a fast, accurate, low-cost tool

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OUR SOLUTION

- **A low-cost, portable Solar Assessment Node that:**

-Measures real-time irradiance, temperature, humidity, GPS
-Uses sensor fusion
-Applies AI/ML to predict energy output & ROI
-Generates site feasibility score instantly
-Uploads everything to a cloud dashboard with maps & analytics

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SYSTEM ARCHITECTURE

1. Sensors
(LDR Array, DHT22, GPS)

2. ESP32

WiFi

3. Cloud Database

4. AI/ML Prediction
Engine

5. Web Dashboard
(Live)

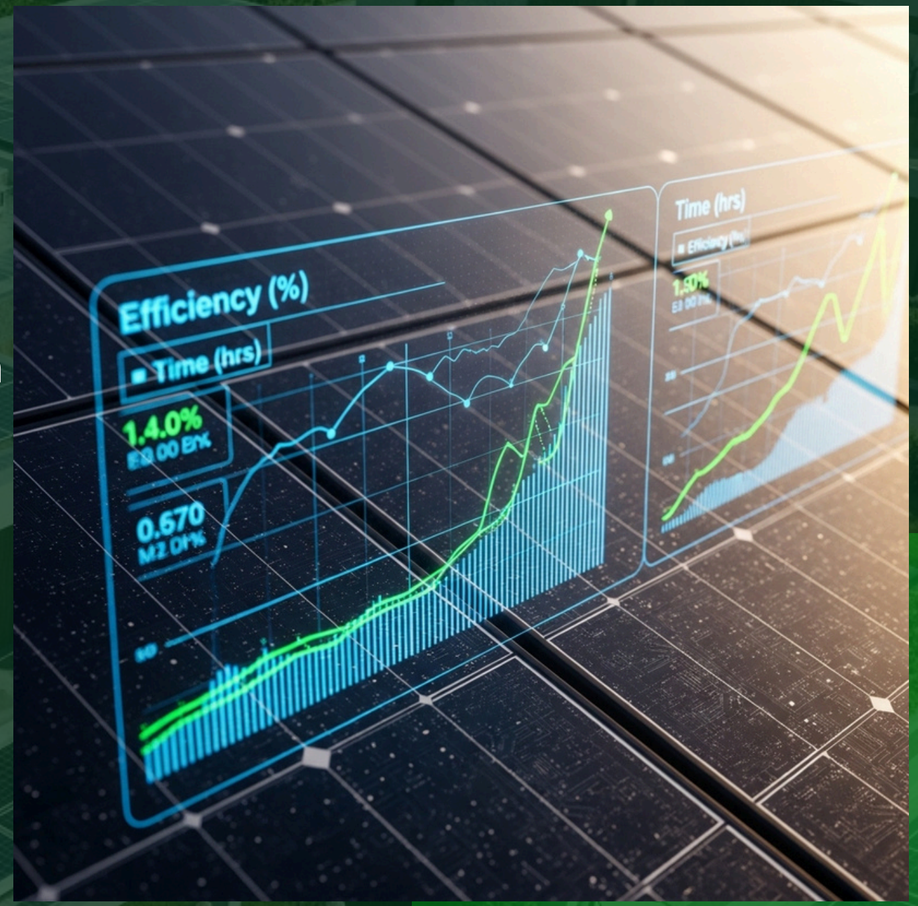
6. Energy Output
ROI
Feasibility

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HOW WE'RE UNIQUE?

No other existing tool combines all of these:

- ✔ Real-time on-site measurement
- ✔ AI-based prediction using hybrid sensor + NASA data
- ✔ Geo-tagged dataset creation
- ✔ Instant ROI & payback analysis
- ✔ Fully portable (power bank operated)
- ✔ Low cost (<12,000LKR)
- ✔ Scalability for national solar mapping



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INNOVATIVE FEATURES



Sensor Fusion

- Multi-LDR array
- Temperature & humidity fusion
- GPS

AI/ML Integration

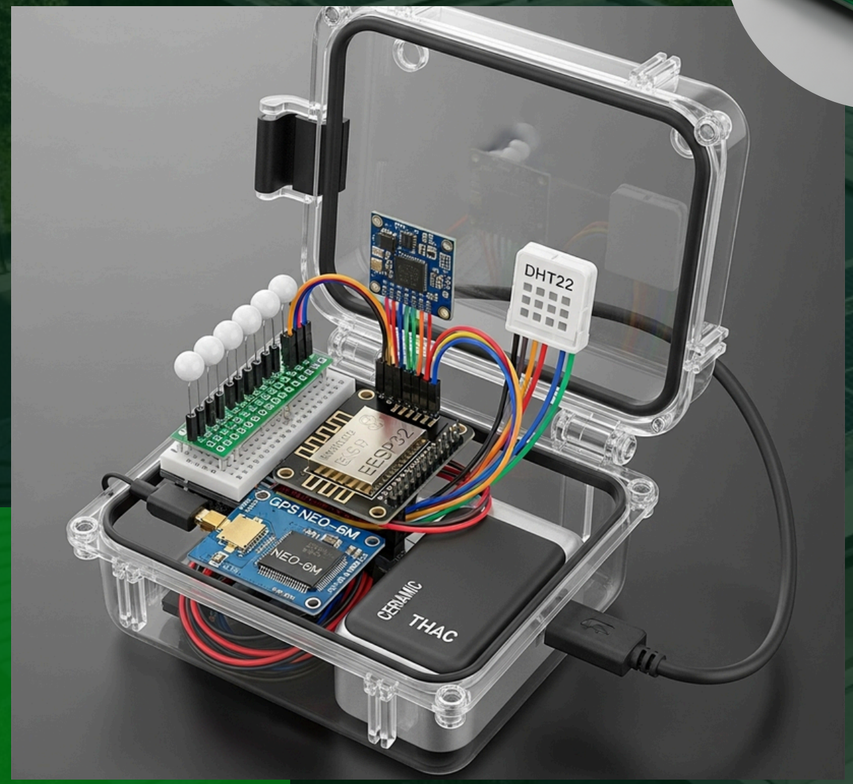
- LightGBM / ANN prediction
- Learns from NASA POWER + local sensor data
- Converts short-term readings → long-term predictions

Portable Solar Intelligence

- Quick deployment anywhere
- Works on power bank
- Cloud-synced analytics

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THE HARDWARE

- ESP32 microcontroller
- LDR/Pyranometer array
- DHT22 temperature/humidity
- GPS NEO-6M
- Power bank + waterproof enclosure
- Cost: ~ LKR 11,500

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WHAT USER GETS

- Live irradiance & weather readings
- Predicted daily/annual PV output
- Location’s feasibility rating
- ROI & payback period
- Map of geo-tagged test points
- Historical logs & graphs



Perfect for CEB, investors, installers, researchers

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MARKET & USE CASES

- Utility & Government
  - National solar mapping
  - Grid planning & policy insights
- Solar Installers / Investors
  - Quick site validation
  - Reduce failed installations
- Engineering Edu + Research
  - Renewable energy analytics
  - IoT + ML learning tool
- Communities
  - Empower local solar adoption

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FUTURE WORK

- Build a national-level irradiance dataset from distributed nodes
- Add satellite image fusion for shading detection
- Integrate actual PV panel output sensor for calibration
- AI-based auto-feasibility classification
- Use LoRaWAN for rural long-distance communication
- Develop mobile app for field engineers
- Enable predictive maintenance for existing solar farms

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WHY THIS MATTERS

- Brings affordable energy intelligence to Sri Lanka
- Helps reduce diesel dependence
- Supports SDG 7: Affordable Clean Energy
- Enables data-driven renewable expansion
- Empowers communities, students, utilities & private sector

Converts a simple IoT device into an AI-powered energy consultant

Our device makes solar energy accessible, intelligent, and optimized for everyone.

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THANK YOU

Eureka

E U R E K A - W A V E